

Data Preprocessing Foundations

Tidy data, missing values, and transformation



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Continuing into Week 2

- Lab 02 begins in Class B and builds on your Lab 01 workflow.
- Use R or Python, locally or in an approved cloud environment.
- Keep assessed work private until the deadline and instructor release.
- Check the course repository for current practical instructions.

Bring your dataset, its data dictionary, and a record of the decisions you made in Lab 01.

From Summaries to Preparation

Last class described the values in a variable. Today we check what those values mean, organize them, and decide how to handle gaps.



One Replacement Can Change the Mean

Ten people have an average income of \$50,000. Replace the highest earner with someone earning far more. The mean rises, while the two middle incomes stay unchanged.

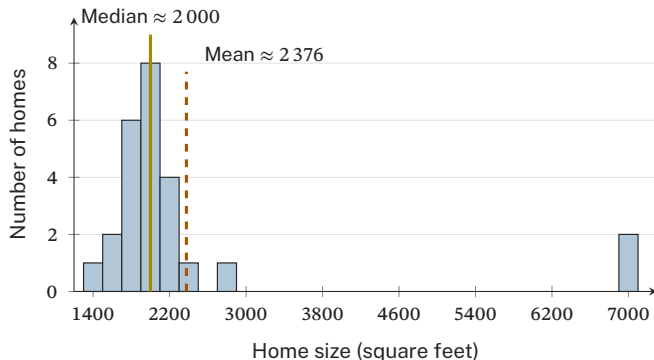
Hypothetical example from Class 02.



What happens to the median in this example?

The mean depends on every magnitude. The median depends on the middle ordered values.

A Long Right Tail Moves the Mean



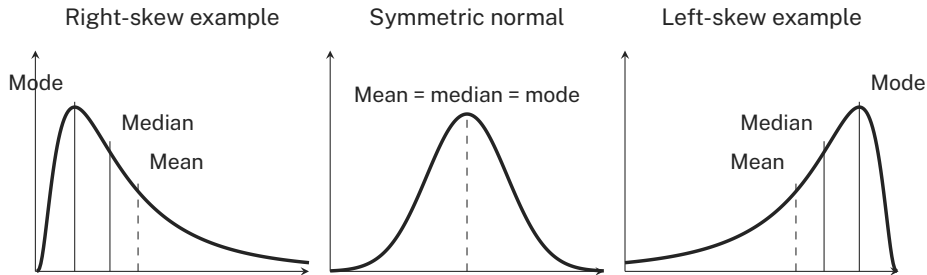
Twenty-five homes.

Twenty-three lie between 1 400 and 2 800 square feet; two are near 7 000.

Bin midpoints give a median of about 2 000 and a mean of about 2 376.

Teaching histogram reconstructed from the original lecture's bin counts. Exact unbinned values are unavailable.

Skew and the Ordering of Summaries



These examples show a common ordering under skew. It is not universal. The normal example is symmetric.

Which Person Is Replaced?

Replace the named earner with someone whose income exceeds every original income.

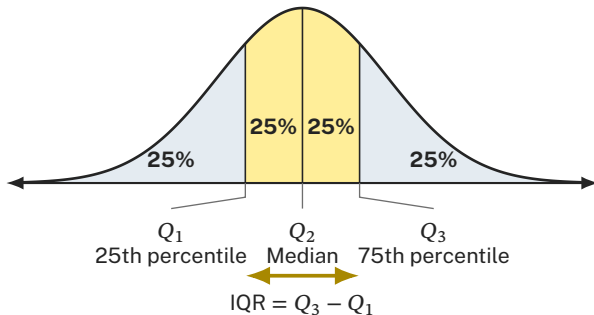
Four incomes	Original median	Lowest replaced	Highest replaced
50k, 50k, 50k, 50k	50k	50k	50k
40k, 40k, 60k, 60k	50k	60k	50k
30k, 40k, 60k, 60k	50k	60k	50k

The median stays unchanged when the middle values stay unchanged. Ties can preserve it even when a low earner is replaced.

Location, Spread, and Potential Outliers

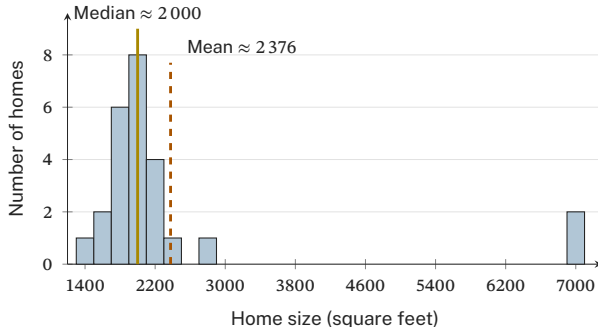
- Location, mean, median, mode.
- Spread, variance, SD, IQR, range.
- Potential outliers, values outside the 1.5 IQR fences.

A flag starts an investigation; it does not establish an error.



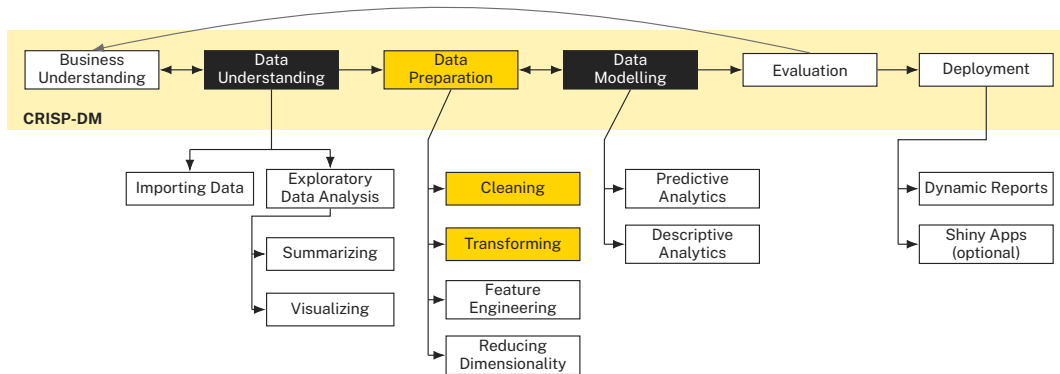
Normal example; both tails extend without bound.

Choose a Summary and Explain It



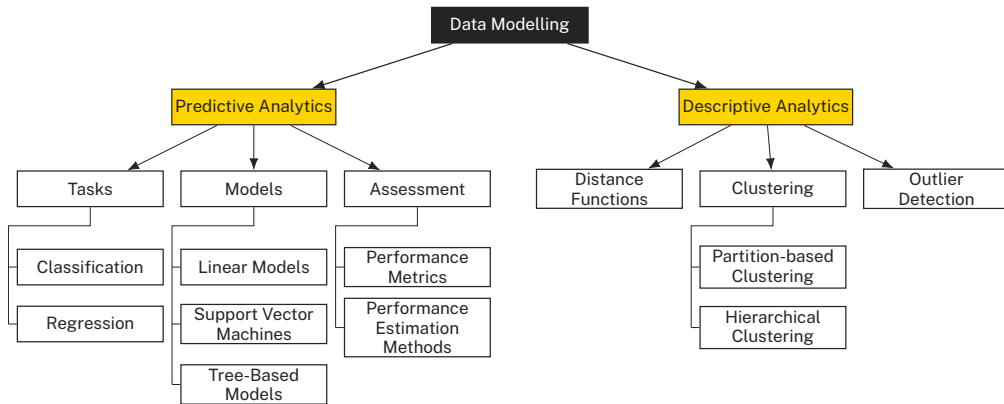
Which would you report for a typical home, mean or median? What would your choice leave out?

Today, Data Preparation



Cleaning and transforming build on the summaries from Class 02.

Later Stages Give Preparation a Purpose



Orientation only. Modelling and model evaluation are taught in later weeks.

What You Will Practise

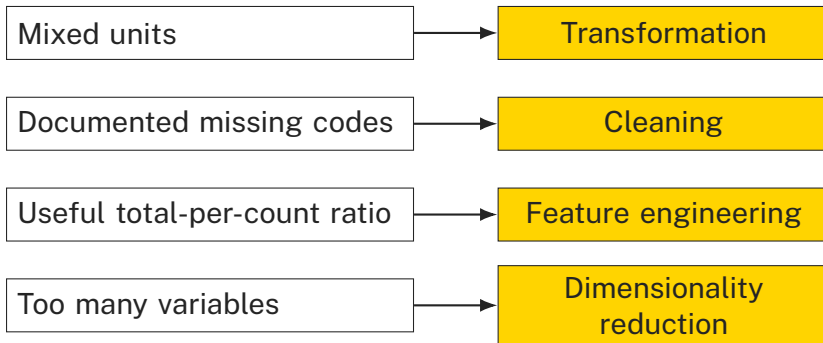
- Connect a data-quality problem to a preparation step.
- Define an observational unit and reshape a table accordingly.
- Distinguish missingness assumptions and compare handling strategies.
- Calculate standardization and binning in R or Python.

Activities alternate between worked examples, individual predictions, and open class discussion.

Why Pre-Process?

- A temperature column mixes Celsius and Fahrenheit. Convert to one unit.
- A code such as 9999 represents a missing value. Decode it using the data dictionary.
- A total and a count can produce a useful new variable, such as amount per observation.
- Too many variables can make analysis difficult. Later methods can reduce them.

Match Each Problem to an Action



Four Preparation Categories

Cleaning	Interpret types, missing codes, inconsistencies, and errors.
Transformation	Change units, scale, or representation.
Feature engineering	Derive useful variables from recorded information.
Dimensionality reduction	Reduce the number of variables used in an analysis.

Today focuses on cleaning and transformation. Class B extends these ideas to feature engineering and introduces dimensionality reduction.

Check Meaning Before Changing a Value

What does one row represent? Which units and types are expected? Which missing codes were documented?

—

A missing value can remain missing. Filling every gap is not the goal.

Start with the Observational Unit

- One variable per column.
- One observation per row.
- One value per cell.
- One type of observational unit per table.

variables

one value

one row

For the next example, an observation is **one student in one course**.

Organize the Grades Table

We want one row for each student-course pair. Which columns need to change?

Think for two minutes, compare with a neighbour, then explain one change.

The example's data dictionary says grades range from 0 to 100 and 9999 means missing.

Name	Math	English	DegreeYear
Luis	90	86	CS 2020
Maria	85	9999	Math 2013
Jose	70	60	CS 2015

Three Decisions, with Different Reasons

Name	Math	English	DegreeYear
Luis	90	86	CS 2020
Maria	85	9999	Math 2013
Jose	70	60	CS 2015

1. Split DegreeYear into two variables.
2. Reshape course names into a Course column and scores into Grade.
3. Replace the documented missing code with NA or NaN.

One Row per Student and Course

Name	Course	Grade	Degree	Year
Luis	Math	90	CS	2020
Maria	Math	85	Math	2013
Jose	Math	70	CS	2015
Luis	English	86	CS	2020
Maria	English	NA	Math	2013
Jose	English	60	CS	2015

Three student rows become six student-course rows. The raw table is retained. Maria's missing English grade remains a missing value.

A student-level analysis could instead treat Math and English grades as separate variables.

Reshape the Grades in R

Continue from the runnable setup in the lecture notes; it creates `clean_grades`.

```
tidy_grades <- tidyr::separate_wider_delim(  
  clean_grades, DegreeYear, delim = "|",  
  names = c("Degree", "Year")  
)  
tidy_grades$Year <- as.integer(tidy_grades$Year)  
tidy_grades <- tidyr::pivot_longer(  
  tidy_grades, c(Math, English),  
  names_to = "Course", values_to = "Grade",  
  cols_vary = "slowest"  
)  
tidy_grades <- tidy_grades[c("Name", "Course", "Grade", "Degree", "Year")]  
print(tidy_grades)  
#
```

Reshape the Grades in Python

Continue from the runnable setup in the lecture notes; it creates `clean_grades`.

```
tidy_grades = clean_grades.copy()
tidy_grades[["Degree", "Year"]] = (
    tidy_grades["DegreeYear"].str.split("|", expand=True, regex=False)
)
tidy_grades["Year"] = tidy_grades["Year"].astype(int)
tidy_grades = tidy_grades.drop(columns="DegreeYear").melt(
    id_vars=["Name", "Degree", "Year"],
    var_name="Course", value_name="Grade",
)
tidy_grades = tidy_grades[["Name", "Course", "Grade", "Degree", "Year"]]
print(tidy_grades)
#
```

Dates and Strings Need Explicit Rules

Normalize labels only when the data dictionary says case and surrounding spaces carry no meaning.

R

```
labels <- c(" North ", "north", "SOUTH")
trimws(tolower(labels))
# "north" "north" "south"
as.Date(c("2026-09-15", "2026-09-16"),
        format = "%Y-%m-%d")
# Calendar dates; timestamps also need an explicit
#   ↪ time zone.
```

R also offers `stringr` and `lubridate`; Python offers string methods and timezone-aware datetime tools. Use either language.

Python

```
labels = pd.Series([" North ", "north", "SOUTH"])
print(labels.str.strip().str.lower())
# "north", "north", "south"
pd.to_datetime(["2026-09-15", "2026-09-16"],
               ↪ format="%Y-%m-%d")
# Calendar dates; timestamps also need an explicit
#   ↪ time zone.
```

A Gap Needs an Explanation

First record what is missing. Then ask how the gap could have arisen and what information supports that explanation.



Three Assumptions About Missingness

MCAR Missingness is independent of both observed and unobserved values.

MAR After accounting for observed information, missingness does not depend on the unobserved values.

MNAR Missingness still depends on unobserved information after accounting for what was observed.

These describe the recording process, not whether the measurements themselves are random.

Hypothetical Examples

	Instrument example	Survey example
MCAR	Randomly selected readings are lost independently of measured values and recording order.	A random subset of forms is lost independently of responses.
MAR	Failures depend on recorded surface type; within each type they do not depend on the reading.	Response depends on recorded delivery method; within that method it does not depend on the answer.
MNAR	An unrecorded extreme reading makes failure more likely.	Response still depends on the undisclosed answer after accounting for recorded variables.

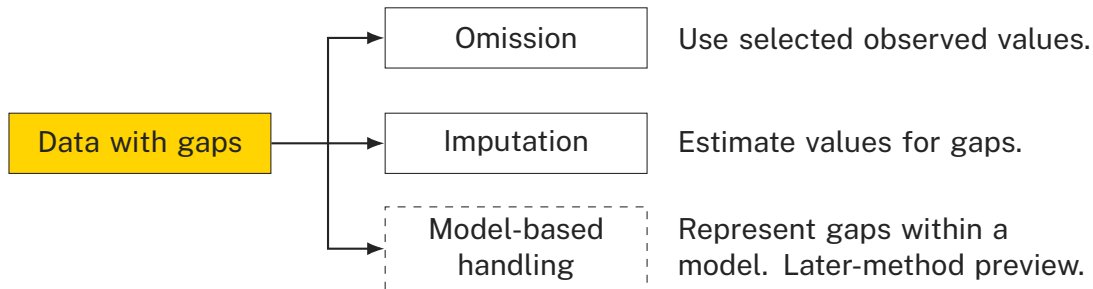
What Evidence Supports the Assumption?

A sensor fails more often outdoors than indoors. Both location and readings are recorded when available.

- What extra assumption would make this MAR?
- What if failure also depends on an unrecorded, extreme sensor reading?

We can look for observed associations, but the observed data alone cannot distinguish MAR from MNAR.

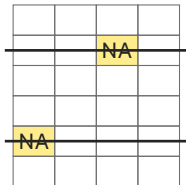
Three Ways to Proceed



Complete Cases and Available Cases

Complete-case deletion

Use only rows observed on every variable needed for this analysis.



		NA	
NA			

Two incomplete rows omitted

MCAR does not add selection bias to a mean from a representative sample; fewer observations still mean less information.

Available-case summaries

Use every observed value needed for each summary. A two-variable statistic uses complete pairs, hence *pairwise* deletion.

Different summaries may describe different subsets. Always report their counts.

Keep the Sample Count Beside the Mean

Use only the two score columns in `clean_grades`.

R

```
scores <- clean_grades[c("Math", "English")]
complete <- scores[complete.cases(scores), ]
nrow(complete) # 2
colMeans(complete)
# Math 80, English 73
colMeans(scores, na.rm = TRUE)
# Math 81.66667, English 73
colSums(!is.na(scores))
# Math 3, English 2
#
```

R needs `na.rm=TRUE`; pandas means skip missing values by default. Report three Math scores and two English scores.

Python

```
scores = clean_grades[["Math", "English"]]
complete = scores.dropna()
print(len(complete)) # 2
print(complete.mean())
# Math 80, English 73
print(scores.mean())
# Math 81.66667, English 73
print(scores.notna().sum())
# Math 3, English 2
#
```

Ways to Fill a Gap

Central value	A mean, median, or mode, selected for the variable type.
Ordered measurements	Previous value (LOCF), baseline value (BOCF), or interpolation between observations. Requires a justified time-series assumption.
Prediction	Other recorded variables inform a predicted numeric value or category. Conceptual preview.
Multiple imputation	Several plausible completed datasets and a combined analysis that represents uncertainty. Conceptual preview.

Imputed values are estimates. Retain a record of which entries were originally missing.

Compare Mean and Median Fill in R

```
observed <- c(2, 4, 4, 5, 7, 9, 12, 41)
with_gaps <- c(observed, NA, NA)
mean_filled <- with_gaps
mean_filled[is.na(mean_filled)] <- mean(observed)
median_filled <- with_gaps
median_filled[is.na(median_filled)] <- median(observed)
c(mean(mean_filled), sd(mean_filled)) # 10.5, 11.22497
c(mean(median_filled), sd(median_filled)) # 9.6, 11.3842
#
```

Compare Mean and Median Fill in Python

```
observed = pd.Series([2, 4, 4, 5, 7, 9, 12, 41], dtype=float)
with_gaps = pd.concat([observed, pd.Series([np.nan, np.nan])], ignore_index=True)
mean_filled = with_gaps.fillna(observed.mean())
median_filled = with_gaps.fillna(observed.median())
print(mean_filled.mean(), mean_filled.std()) # 10.5, 11.22497
print(median_filled.mean(), median_filled.std()) # 9.6, 11.3842
#
```

Predict the Effect of Mean Fill

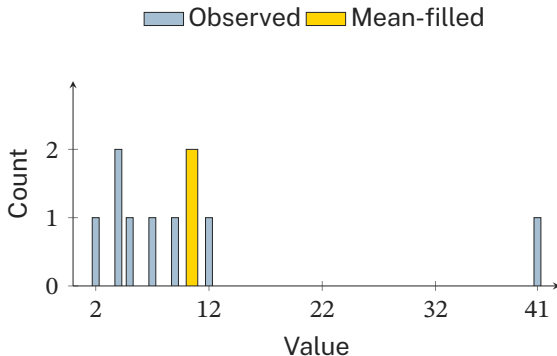
Replace two gaps with the observed mean, 10.5. What happens to the mean and sample standard deviation?



Write a prediction and a reason before calculating.

The Mean Stays, the Spread Shrinks

- Preserves the observed mean exactly.
- Reduces the sample variance by adding zero deviations.
- Can distort relationships between variables.
- Does not remove selection bias in the observed values.



Calculate the Change in Variance

Observed values ($n = 8$)

$$\bar{x} = 84/8 = 10.5$$

$$\sum_i (x_i - \bar{x})^2 = 1134$$

$$s_x^2 = 1134/7 = 162$$

$$s_x = \sqrt{162} \approx 12.73$$

Two mean-filled gaps ($n = 10$)

$$\bar{x} = 105/10 = 10.5$$

$$1134 + 0 + 0 = 1134$$

$$s_x^2 = 1134/9 = 126$$

$$s_x = \sqrt{126} \approx 11.22$$

The mean remains 10.5 under any missingness mechanism. MCAR concerns whether this observed mean estimates the population mean without selection bias.

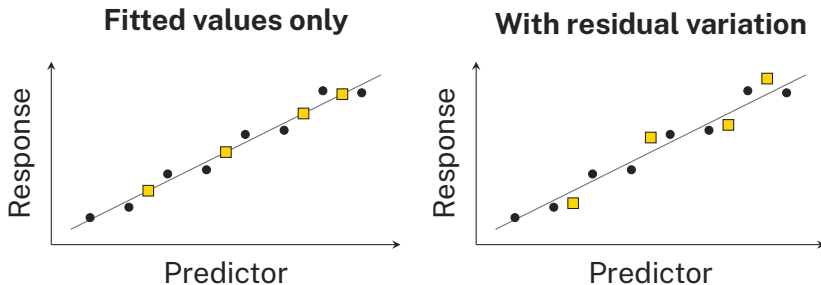
SD falls about 11.8% relative to the observed-data SD. The true complete-data SD is unknown.

Regression Imputation

A regression model predicts a numeric value from other recorded variables.
Fitting is taught later.

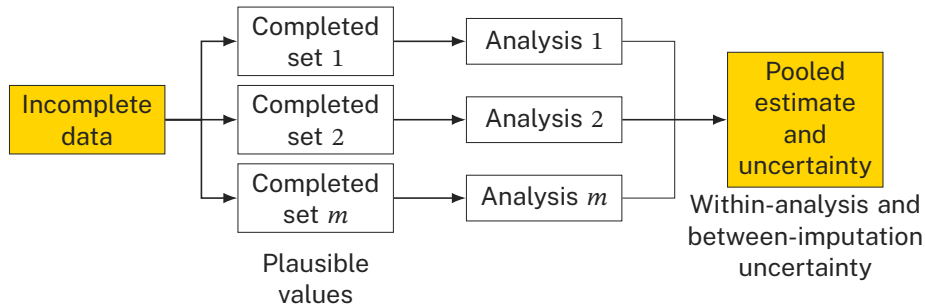
Potential benefit	Limitation
Uses other recorded information.	Needs a suitable model and observed predictors.
Predicts an expected numeric value.	Does not recover the unknown individual value.
Can improve on a single overall mean.	Deterministic predictions omit residual variation and can exaggerate associations.

Predictions Need Uncertainty



Circles are observed values; gold squares are fitted values. Left, fitted values omit residual variation. Right, one illustrative draw adds residual variation. A suitable model and uncertainty analysis are still needed.

Multiple Imputation



Analyze m plausible completed datasets, then combine estimates and uncertainty. Standard applications assume MAR and suitable models. MNAR requires additional assumptions and sensitivity analysis.

Tools Used After Model Fitting Is Taught

R

`recipes` can organize imputation steps.
`mice` creates completed datasets, and
`pool` combines fitted-model results.

Nearest-neighbour imputation uses similar records. Choosing distances, models, and pooling rules is later material, not required for Lab 02.

Python

`SimpleImputer` supplies a stated rule.
`Statsmodels MICE` supports repeated imputation and pooled analysis.
`IterativeImputer` alone is not pooled analysis.

Defend a Missing-Data Decision

A variable has two missing measurements. Would you report its observed values, fill the gaps, or seek more information?

- State the question and the available evidence.
- Name your assumption and the limitation it creates.
- Explain how you would record the decision.

New Representation, Same Meaning

With units, variable types, and missingness decisions documented, choose a representation that suits the analysis.

Today, standardization and discretization.

Express Values in Sample-SD Units

$$z_i = \frac{x_i - \bar{x}}{s_x}$$

Use a numeric variable with at least two observed values and $s_x > 0$.

Constant columns need a separate decision; do not divide by zero.

For 2, 4, 4, 5, 7, 9, 12, 41, $\bar{x} = 10.5$ and $s_x \approx 12.73$.

$z_1 \approx -0.67$, $z_4 \approx -0.43$, $z_8 \approx 2.40$.

The mean becomes 0 and sample SD becomes 1. Shape and ordering remain unchanged.

Use the Same Sample-SD Convention

R

```
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
stopifnot(length(x) > 1, sd(x) > 0)
z <- (x - mean(x)) / sd(x)
z[1] # -0.6678231
mean(z) # approximately 0
sd(z) # 1
as.vector(scale(x))[1] # -0.6678231
#
```

Python

```
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41], dtype=float)
assert len(x) > 1 and x.std(ddof=1) > 0
z = (x - x.mean()) / x.std(ddof=1)
print(z.iloc[0]) # -0.6678231
print(z.mean()) # approximately 0
print(z.std(ddof=1)) # 1
# Both examples use the sample SD.
#
```

NumPy's default SD and scikit-learn's StandardScaler use divisor n . This course example uses $n - 1$ in both languages.

Equal Widths or Quantile Boundaries?

Equal-width bins split the numeric range into k intervals. Quantile bins aim for similar counts, but ties can prevent equal counts.

Equal-width bins



Quantile bins; ties retain the same category



Numbers beneath intervals are counts; bottom labels show the value scale.

Stacked marks at 4 represent the two equal observations.

Save the Boundaries in R

```
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
k <- 4
width_edges <- seq(min(x), max(x),
                    length.out = k + 1)
quantile_edges <- quantile(
  x, probs = seq(0, 1, 1/k))
table(cut(x, width_edges,
          include.lowest = TRUE))
# 6, 1, 0, 1
table(cut(x, quantile_edges,
          include.lowest = TRUE))
# 3, 1, 2, 2
#
```

Include the minimum explicitly. Reuse saved edges for comparable categories.

Save the Boundaries in Python

```
x = pd.Series([2, 4, 4, 5, 7, 9, 12, 41])
k = 4
width_edges = np.linspace(
    x.min(), x.max(), k + 1)
quantile_edges = x.quantile(
    np.linspace(0, 1, k + 1))
print(pd.cut(x, width_edges,
             include_lowest=True).value_counts(sort=False))
# 6, 1, 0, 1
print(pd.cut(x, quantile_edges,
             include_lowest=True).value_counts(sort=False))
# 3, 1, 2, 2
#
```

If quantile boundaries repeat, use fewer bins and document the choice.

Age Non-Disclosure

A hypothetical survey records age and consumption. Respondents under 21 are less likely to disclose age, even at the same consumption level. No other variables are recorded.

- What does this suggest about the missingness mechanism?
- Could age categories recover ages that were never reported?
- What can a separate “age not disclosed” flag tell us?

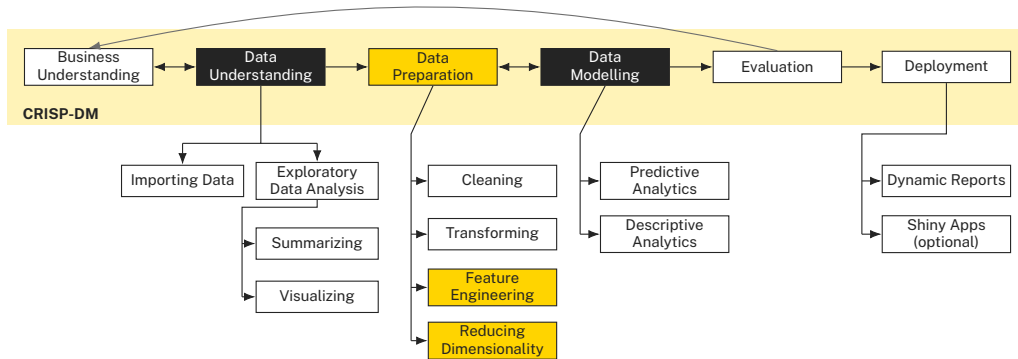
Keep unknown ages unknown. Without evidence, do not label a nonrespondent “under 21.”

What to Take Away

- Define the observational unit before reshaping.
- Use the data dictionary to interpret types, units, and missing codes.
- Report missing counts and justify any deletion or imputation.
- Standardization changes scale; binning reduces detail.
- Keep raw data and record every transformation.

Model-based imputation is a conceptual preview; the practical uses today's guided foundations.

Extend the Preparation Workflow



Class B continues cleaning and transformation, adds derived variables, and begins Lab 02.

Readings and Optional Extensions

- Wickham, *Tidy Data* (2014). Observations, variables, and table structure.
- `tidyr`, `pivot_longer` and `pandas`, `melt`. Read the examples for your chosen language.
- van Buuren, *Flexible Imputation of Missing Data*, Section 1.2. Missingness mechanisms.
- Optional extension, Section 1.3 on imputation trade-offs and the `mice` pooling workflow. Model fitting and pooling are not required for Lab 02.